Applied SVAR Identification

Day 1: VAR Foundations

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Monday 22 June 2026

Administration

- **Dates:** Monday 22 June – Friday 26 June 2026
- **Time:** 10:00–12:30 each day (12.5 lecture hours total), including a 10-minute break
- **Format:** Integrated discussion of essential theory with applications
- **Applied Approach:** Focus on key ideas, implementation, and interpretation; for theory see Hamilton (1994) or Kilian & Lütkepohl (2017)
- **Software:** MATLAB Toolbox by Ambrogio Cesa-Bianchi and some ad hoc coding

Course Roadmap

Day	Topic
1	VAR Foundations: Motivation, Estimation, Specification, Identification & Tools
2	Identification I: Short-Run Restrictions
3	Identification II: Long-run Restrictions
4	Identification III: Sign, Magnitude, and Narrative Restrictions
5	Identification IV: Instruments

Key Applications

Day	Papers
1	Stock and Watson (2001)
2	Kilian (2009); Wong (2015);
3	Blanchard & Quah (1989); Bjørnland (2009)
4	Uhlig (2005); Kilian & Murphy (2012); Antolín-Díaz & Rubio-Ramírez (2018)
5	Gertler & Karadi (2015); Känzig (2021); Paul (2020)

Learning Objectives

By the end of today, you should be able to:

1. Explain why VAR models are a natural tool for macroeconomic and policy analysis.
2. Distinguish between a reduced-form VAR and a structural VAR (SVAR).
3. Specify a VAR model for a given application.
4. Use a VAR for macroeconomic forecasting.
5. Understand the identification problem.
6. Implement and interpret SVAR tools: impulse responses, variance decompositions, and historical decompositions.
7. Estimate the Stock and Watson (2001) stylized SVAR model of the macroeconomic effects of monetary policy in MATLAB.

Part 1

Why VARs?

"You can't know where you are going until you know where you have been." — Maya Angelou

The Beginnings of Macroeconomics (and Macroeconometrics)

1929–1939	Wall Street Crash & Great Depression: Economics criticized for lacking general theory and predictive tools
1930	Econometric Society founded by Frisch, Fisher, Schumpeter and others — Cowles provided funding for the society and Econometrica
1932	Cowles Commission established to put economic modeling on a scientific footing: from *Science is Measurement* to *Theory and Measurement*
1933	Econometrica launched with Frisch as Editor
1936	Keynes, *General Theory*: Laid the foundations for macroeconomics
1920s-1940s	Frisch, Slutsky, Haavelmo (and others): Incorporated probability and general principles of statistical inference into econometrics
1950s-1970s	Klein–Goldberger models: Large dynamic simultaneous equations models (DSEMs) of the US with many more equations than variables (see Fox (1956))

Two Problems with Models Before the 1980s

1. **The Lucas critique** *Lucas (1976)*
   - We can't use DSEMs with historical data to predict the future effects of a policy change, because the policy change itself alters people's behaviour.
   - A model estimated under one policy regime cannot reliably predict behaviour under a different regime.
   - **His proposal:** Specify models with microfoundations (eventually DSGE)!
2. **Incredible restrictions** *Sims (1980)*
   - Two types of incredible restrictions in DSEMs were largely unjustifiable from theory:
     1. Treating some variables as exogenous.
     2. Constraining the dynamics of the endogenous variables and of the error term.
   - **His proposal:** Treat all macroeconomic variables symmetrically, let the data determine the dynamics, and impose as few restrictions as necessary $\rightarrow$ VAR!

Key Idea

Together, these critiques undermined confidence in DSEMs as a useful model. Here we focus on VARs!

A Unified Framework

Stock and Watson (2001) summarise the task of macroeconometrics into 4 tasks:

1. Describe and summarise macroeconomic data;
2. Make macroeconomic forecasts;
3. Quantify what we do or do not know about the true structure of the macroeconomy;
4. Advise (and sometimes become) macroeconomic policymakers.

VARs can do all 4 tasks!

Here we illustrate task 2 (forecasting) and focus on task 3 (structural analysis).

Part 2

The VAR Model

Notation and structure

What Are VAR Models?

- Can model multiple time series simultaneously within a single model
  - Example: GDP, inflation & interest rates are all interdependent

- Most commonly used model in applied macroeconomics (underpin DSGE models)

- Come in two types
  1. **Reduced form VARs** are the benchmark model for **forecasting**
     - Example: what do we expect the interest rate to be next quarter, conditional on knowing GDP and inflation today?

  2. **Structural VARs** are the foundational model for **structural analysis**
     - Example: How do we expect inflation and output to respond to a 25 basis point increase in the interest rate?

Motivating Example: Two variable SVAR(1) model

Suppose we have the following system of equations of two variables (e.g., inflation and interest rate):

$$\begin{aligned} y_{1,t} &= 1 + 0.6y_{2,t} + .5y_{1,t-1} + \varepsilon_{1,t} \\ y_{2,t} &= 1 + 0.4y_{1,t} + y_{1,t-1} + .2y_{2,t-1} + \varepsilon_{2,t} \end{aligned} \tag{1}$$

We can take all **contemporaneous terms** (those with date t) to the left hand side and write it using matrix notation

$$\begin{bmatrix} 1 & -0.6 \\ -0.4 & 1 \end{bmatrix} \begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} .5 & 0 \\ 1 & .2 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{bmatrix} \tag{2}$$

Or more compactly as

$$B_0 y_t = b + B_1 y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, I) \tag{3}$$

We refer to this as a **structural VAR model** of order 1, denoted SVAR(1).

General SVAR model

The general SVAR with p lags and n variables can be written as

$$B_0 y_t = b + \sum_{j=1}^{p} B_j y_{t-j} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}\left(0, I\right) \tag{4}$$

where

- y_t is a $n \times 1$ vector of variables of interest
- b is a $n \times 1$ vector of constants
- B_j, $j = 1, ..., p$, are $n \times n$ matrices of autoregressive coefficients
- ε_t is a $n \times 1$ vector of **structural shocks**.
- B_0 is an $n \times n$ matrix that shows the contemporaneous relationships between variables at the current date t.

What is a Structural Shock?

A **structural shock** ε_t is a primitive exogenous, orthogonal, and economically interpretable disturbance that moves the economy (Ramey, 2016).

Key properties:

- *Exogenous* — not caused by other variables in the system at time t
- *Orthogonal* — uncorrelated with all other structural shocks: $\mathbb{E}[\varepsilon_t \varepsilon_t'] = I$
- *Interpretable* — each shock has a clear economic meaning

Example from Stock & Watson (2001):

- $\varepsilon_{1,t}$: inflation shock (e.g. aggregate supply disturbance)
- $\varepsilon_{2,t}$: unemployment shock (e.g. aggregate demand)
- $\varepsilon_{3,t}$: interest rate shock (e.g. monetary policy)

What do Structural Shocks Allow us to do?

Once we have identified structural shocks ε_t, three key tools become available:

1. **Impulse Response Functions (IRFs)**
   - Quantify the dynamic response of each variable to a one-unit structural shock
   - Example: how does inflation evolve following a monetary policy tightening?
2. **Forecast Error Variance Decompositions (FEVDs)**
   - Quantify the variation in each variable from each shock
   - Example: how much of the variation in inflation is driven by monetary policy shocks vs. aggregate supply vs. aggregate demand shocks?
3. **Historical Decompositions (HDs)**
   - Attribute historical fluctuations in each variable to each structural shock
   - Example: how did monetary policy shocks contribute to the Great Recession?

Simultaneity Problem

Problem: The SVAR suffers from **simultaneity**, i.e. a change in one variable contemporaneously generates changes in other variables. This results in the OLS estimator becoming inconsistent.

Solution: Write the SVAR as a reduced form VAR and estimate that using OLS. Then recover the SVAR for structural analysis.

From SVAR to reduced form VAR

Inverting B_0 allows us to write the SVAR as a reduced form VAR (VAR or VAR):

$$y_t = c + \sum_{j=1}^{p} A_j y_{t-j} + e_t, \quad e_t \sim \mathcal{N}(0, \Sigma) \tag{5}$$

where

- $c = B_0^{-1} b$ is a $n \times 1$ vector of constants
- $A_j = B_0^{-1} B_j$, $j = 1, ..., p$, are $n \times n$ matrices
- $e_t = B_0^{-1} \varepsilon_t$ is a $n \times 1$ vector of error terms

The reduced form covariance matrix is

$$\Sigma = \mathbb{E}\left[e_t e_t^{'}\right] = B_0^{-1} \left(B_0^{-1}\right)' \tag{6}$$

We call B_0^{-1} the **impact matrix** since it governs the instantaneous response of y_t when ε_t changes. The goal of this course is to recover this matrix!

The SVAR is Underidentified

Objective: Estimate the VAR and then recover the SVAR.

Identification Problem: The SVAR has more parameters than the estimated reduced form VAR resulting in an **underidentified system of equations**. Check:

- SVAR has
  - $k = n(np + 1)$ autoregressive parameters
  - n^2 contemporaneous parameters
- VAR has
  - $k = n(np + 1)$ autoregressive parameters
  - $\frac{n(n+1)}{2}$ (unique!) parameters in the covariance matrix

Key Idea

Identification problem: The SVAR is not identified. Extracting the SVAR from the VAR therefore requires identifying assumptions — the main focus of this course!

Part 3

Estimation

OLS and bootstrapping

Running Application: Stock and Watson (2001)

Stock and Watson (2001, JEP) consider a three-variable quarterly US VAR with:

	Variable	Description
π_t	Inflation	GDP deflator inflation (annualised, %)
u_t	Unemployment	Unemployment rate (%)
R_t	Fed Funds Rate	Federal funds rate (%)

- Quarterly data from 1960:Q1–2001:Q3.
- Four lags

Stock and Watson (2001): The Data

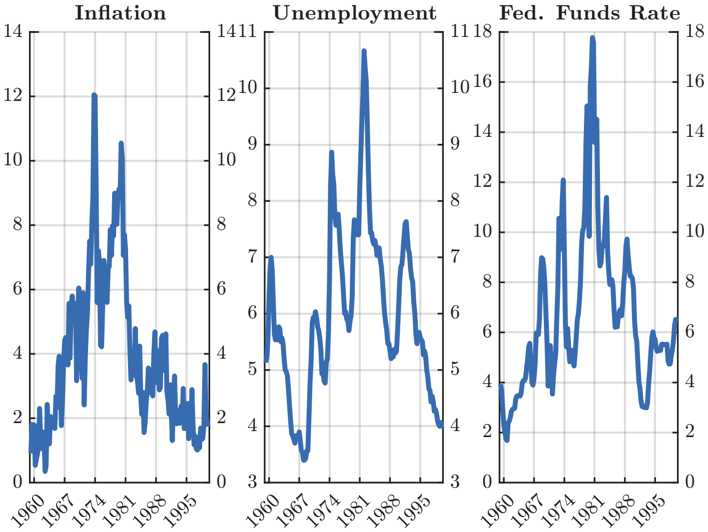


Figure: Inflation, unemployment rate, and Federal funds rate from 1960:Q1–2001:Q3.

OLS Estimation

Each equation of the VAR has the same regressors:

$$y_{it} = c_i + a'_{i1} y_{t-1} + \cdots + a'_{ip} y_{t-p} + e_{it}, \quad i = 1, \ldots, n.$$

Key result: applying OLS equation-by-equation is consistent and efficient under standard conditions (the Gauss–Markov setup applies in each equation separately).

For the SW (2001) VAR:

- $n = 3$ variables, $p = 4$ lags, $T \approx 162$ quarterly observations.
- Each equation has $np + 1 = 13$ parameters.
- Total parameters: $n^2 p + n = 45$.

Bootstrapping for Inference

OLS standard errors are used for reduced form coefficients, but for structural tools (more on this later) the toolbox uses **bootstrapped confidence bands**.

Basic Residual Bootstrap

1. Estimate the VAR by OLS; collect residuals $\hat{u}_1, \ldots, \hat{u}_T$.
2. Draw T residuals with replacement to create a new residual series.
3. Simulate a new dataset by running the estimated VAR forward with the re-sampled residuals.
4. Re-estimate the VAR on the simulated dataset; compute the statistic of interest.
5. Repeat 500–1000 times; use the empirical quantiles as confidence bands.

Key Idea

We bootstrap to get confidence intervals for the estimates.

MATLAB: Estimating a VAR

Approach	Pros	Cons
Code yourself	Full control; understanding	Time-consuming; error-prone; reinventing the wheel
Econometrics Toolbox	Tested, documented & maintained by MathWorks	Limited flexibility; Difficult to modify
Others' code	No need to code manually; feature-rich	Error-prone; may lack documentation; no guarantees for backward compatibility

Key Idea

Here we use the ACB Toolbox. It's free, well-documented, regularly updated, and covers all of the main identification schemes for SVARs.

Part 4

Specification

Variables, transformations, lags, and stability

Key Specification Questions

1. Which variables?
2. What transformations?
3. How many lags?

Which Variables?

Two schools of thought:

1. Theory $\rightarrow$ Empirics: Let the question dictate the model
2. Data driven $\rightarrow$ Use the kitchen-sink (big data) and let the data decide.

Here we will follow option 1 since it is most relevant for macroeconomic analysis.

SW (2001) choice

The three series: output/unemployment, inflation and FFR are related through the 3-equation NK model. Typical minimal set of variables for closed economy model.

Why not option 2?

Estimation Problems: Over parameterization

- A VAR(p) model with n variables contains $n(1+np)$ coefficients. In macroeconomics, we often find ourselves in situations where the number of parameters is close to the number of observations and run into degree of freedom problems. Called the **over parameterization problem**.
- Since we cannot include all variables of potential interest, we have to make careful choices by referring to economic theory or *a priori* ideas when choosing variables.
- One simple way to determine the usefulness of a variable is *Granger causality*. Another is to apply shrinkage methods like LASSO or Bayesian estimation (touch on this on Day 5).

Granger Causality

- Main idea: if X causes Y, then X should help predict Y.
- Formally: x_t is said to *Granger cause* y_t if it predicts y_t in a statistically significant way $\Rightarrow$ Granger causality is therefore a predictability test (not true causality)
- Example 1: Reconsider the VAR system in equation (2):

$$A_1 = \begin{bmatrix} .5 & 0 \\ 1 & .2 \end{bmatrix}$$

  Here $y_{1,t}$ Granger causes $y_{2,t}$, but $y_{2,t}$ does not Granger cause $y_{1,t}$. Why?
- Example 2: Consider the coefficient matrix:

$$A_1 = \begin{bmatrix} 0.5 & 0.2 \\ 1 & 0.2 \end{bmatrix}$$

  then both $y_{1,t}$ and $y_{2,t}$ Granger cause each other.

Granger Causality: Formal Test

Question: Does y_{jt} help predict y_{it}, over and above its own lags?

Test: In the equation for y_{it}, jointly test whether all p lags of y_{jt} are zero:

$$H_0 : a_{ij,1} = a_{ij,2} = \cdots = a_{ij,p} = 0$$
$$H_1 : \text{at least one } a_{ij,k} \neq 0$$

using an F-test with p restrictions. Reject $H_0 \Rightarrow y_{jt}$ Granger causes y_{it}.

Granger causality is *predictive*, not structural

A significant result means y_{jt} is a useful predictor for y_{it}. It does *not* mean y_{jt} structurally causes y_{it} in an economic sense.

SW (2001): Granger Causality Results

Table 1A: p-values for the null of no Granger causality from row to column variable; small values indicate rejection.

	Inflation shock	**Unemployment shock**	**Int. Rate shock**
Inflation (π)	0.00	0.31	0.00
Unemployment (u)	0.02	0.00	0.00
Fed Funds Rate (R)	0.27	0.01	0.00

Ex: Interpret the results. Would you exclude any of the variables from the model?

Which Transformations?

Again, let theory $\rightarrow$ empirics!

- However, one challenge is that VAR estimation requires that the VAR model is 'stable'. Otherwise, the tools, e.g., forecasts, IRFs, etc, will explode.
- It is therefore common to model series in growth rates which are usually stationary (e.g., pass an ADF test).
- However level series can also be used as long as the VAR is stable...

Warning

If the VAR is not stable then reconsider the data transformation before proceeding. Typical transformations include logs and differences (growth rates).

When is a VAR Stable?

To check stability we write the VAR(p) model as a VAR(1) model using the **companion form representation**:

$$\begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix} = \begin{bmatrix} c \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & \cdots & I & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p} \end{bmatrix} + \begin{bmatrix} e_t \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tag{7}$$

which can be written as

$$Z_t = \Gamma_0 + \Gamma_1 Z_{t-1} + v_t \tag{8}$$

Stability of the VAR(p) model

The VAR(p) model is stable if all *eigenvalues* of the companion form matrix Γ_1 are all less than one in absolute value.

Stability of VARs cont'd

The eigenvalues of Γ_1 are value of λ for which:

$$\det(\Gamma_1 - \lambda I) = 0$$

- **Example**: VAR(1), $n = 2$, with $\Gamma_1 = A_1 = \begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix}$:

$$\det\left(\begin{bmatrix} 0.5 & 0 \\ 1 & 0.2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \det\left(\begin{bmatrix} 0.5-\lambda & 0 \\ 1 & 0.2-\lambda \end{bmatrix}\right)$$

$$= (0.5-\lambda)(0.2-\lambda) = 0$$

$$\downarrow$$

$$\lambda_1 = 0.5, \quad \lambda_2 = 0.2$$

- Since both eigenvalues are < 1 in absolute value $\Rightarrow$ this VAR is **stable**.

Exercise: Is the SW(2001) model stable? Hint: In the ACB Toolbox, `VAR.Fcomp` is the companion matrix.

How Many Lags?

Three schools of thought:

1. Theory $\rightarrow$ Empirics: Let the question dictate the lags!
2. Data driven $\rightarrow$ Use information criteria
3. Rule of thumb: Use one year's worth of lags, e.g., quarterly data $\rightarrow$ p=4

Here we will follow option 3 since Hamilton and Herrera (2004) show that information criteria can choose small lags that distort economic results.

SW (2001) choice

Four lags ($p = 4$) for quarterly data — a standard choice for quarterly macro VARs capturing up to one year of dynamics.

Part 5

VAR Tools

Forecasting

VAR Forecasting

Once estimated, a VAR can produce **iterated h-step-ahead forecasts** by iterating the model forward:

$$\hat{y}_{T+1|T} = \hat{c} + \hat{A}_1 y_T + \hat{A}_2 y_{T-1} + \cdots + \hat{A}_p y_{T+1-p} \tag{9}$$

$$\vdots$$

$$\hat{y}_{T+h|T} = \hat{c} + \hat{A}_1 \hat{y}_{T+h-1|T} + \cdots + \hat{A}_p \hat{y}_{T+h-p|T} \tag{10}$$

- Forecasts for all n variables are produced simultaneously from a single model.
- VARs are a standard benchmark in central bank forecasting competitions.

Policy Lens

A key advantage of VARs over univariate AR models is that they leverage cross-variable information: the lag of unemployment helps forecast inflation, and vice versa.

Forecasting with VARs: Why Stability Matters

- Focus on the VAR(1) model since any VAR(p) has this representation.
- By **recursive substitution**, the VAR(1) equation at time $t+h$ is:

$$y_{t+h} = (I + A_1 + \cdots + A_1^{h-1})c + A_1^h y_t + \sum_{i=0}^{h-1} A_1^i \, e_{t+h-i}$$

- The **VAR point forecast** for y_{t+h} takes expectations conditional on $\mathcal{I}_t$:

$$\hat{y}_{t+h} = \mathbb{E}[y_{t+h} \mid \mathcal{I}_t] = (I + A_1 + \cdots + A_1^{h-1})c + A_1^h y_t$$

- If the VAR is **stable** then $A_1^h \to 0$ as $h \to \infty$, so the point forecast converges to the unconditional mean:

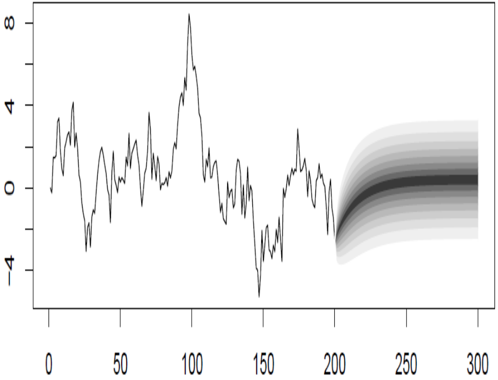
$$\hat{y}_{t+h} \;\to\; (I - A_1)^{-1} c \qquad \text{as } h \to \infty$$

Key Idea

Stability guarantees that long-horizon forecasts settle at the unconditional mean rather than diverging. In contrast, an unstable VAR produces forecasts that explode.

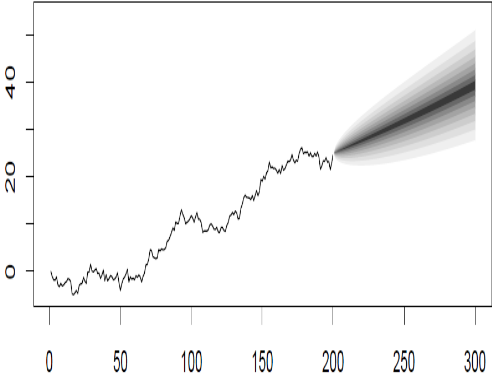
Stable vs Unstable VAR: Point Forecast Convergence

Stable VAR



Point forecasts converge to the unconditional mean as $h \to \infty$

Unstable VAR



Point forecasts diverge to $\pm\infty$, a common sign of misspecification

Evaluating Forecast Accuracy

Step 1: Train/evaluation split. Divide the sample into:

- *Estimation window* $t = 1, \ldots, T_0$ — used to estimate the VAR.
- *Evaluation window* $t = T_0 + 1, \ldots, T$ — held back to assess forecast performance.

Step 2: Forecast. From each origin T_0, produce h-step-ahead forecasts $\hat{y}_{T_0+h|T_0}$.

Step 3: Forecast error. Compare forecast to realisation:

$$e_{T_0+h} = y_{T_0+h} - \hat{y}_{T_0+h|T_0}.$$

Step 4: RMSE. Summarise accuracy across all forecast origins:

$$\text{RMSE}(h) = \sqrt{\frac{1}{M}\sum_{m=1}^{M} e^2_{T_0+m+h}}.$$

Key Idea

A smaller RMSE means better forecast accuracy. Comparing RMSE across models reveals which model best captures the data dynamics.

Stock & Watson (2001): Out-of-Sample Forecast RMSEs

Table 2: Root mean squared errors, 1985:I–2000:IV. Smaller is better.

	Inflation			Unemployment			Interest Rate		
Horizon	RW	AR	VAR	RW	AR	VAR	RW	AR	VAR
2 quarters	0.82	0.70	**0.68**	0.34	**0.28**	0.29	0.79	0.77	**0.68**
4 quarters	0.73	0.65	**0.63**	0.62	**0.52**	0.53	1.36	1.25	**1.07**
8 quarters	**0.75**	**0.75**	**0.75**	1.12	0.95	**0.78**	2.18	1.92	**1.70**

Policy Lens

The VAR matches or beats the AR and random walk at most horizons — particularly for inflation and the interest rate. For unemployment, the AR edges the VAR at short horizons (2 and 4 quarters), while the VAR dominates at 8 quarters.

Part 6

Structural VARs

The identification problem

The Identification Problem

Estimating the VAR gives an estimate of the reduced form covariance matrix:

$$\hat{\Sigma} = B_0^{-1} \left(B_0^{-1} \right)'$$

which has $\frac{n(n+1)}{2}$ unique parameters. E.g., with SW (2001) data:

$$\hat{\Sigma} = \begin{bmatrix} 0.97 & -0.01 & 0.11 \\ -0.01 & 0.05 & -0.09 \\ 0.11 & -0.09 & 0.78 \end{bmatrix}$$

However, the structural impact matrix has n^2 unique parameters. This means that there are $n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}$ unidentified elements.

Key Idea

The SVAR has more parameters than the VAR making it underidentified! Without additional identifying assumptions, there are infinitely many structural decompositions consistent with the data. Identification requires assumptions about the parameters to recover the underlying SVAR.

Common Identification Strategies

Many identification strategies exist. The most commonly used are:

1. Zero Restrictions: Short-run and Long-run
2. Sign, Magnitude, and Narrative Restrictions
3. Instrumental variables
4. Prior Beliefs: Bayesian

New ideas are always welcome!
See Wolf (2020) for a recent discussion of common identification strategies for monetary policy shocks in SVARs.

Identification Strategy 1: Short-run Zero Restrictions

- Sims proposed to identify the SVAR assuming a **recursive system** with parameters above the main diagonal of the impact matrix restricted to be zero:

$$B_0^{-1} = \begin{bmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}$$

- Take as given for now and discuss in more detail tomorrow

SW (2001): Recursive Ordering

SW (2001) follow Sims (1980) and assume a recursive ordering:

$$y_t = (\pi_t,\ u_t,\ R_t)'.$$

Economic interpretation of the restrictions:

1. **Inflation (π_t)** does not respond contemporaneously to unemployment or interest rate shocks. Prices are *sticky*: they are predetermined within a quarter.
2. **Unemployment (u_t)** does not respond contemporaneously to interest rate shocks. Labour market adjustment takes at least one quarter.
3. **Fed funds rate (R_t)** can respond contemporaneously to inflation and unemployment: the Fed reacts to current conditions within a quarter.

Policy Lens

The identification strategy embeds a *theory of monetary transmission*: policymakers react to changes in inflation and output quickly, however monetary policy affects inflation and unemployment with a lag.

The Great Schism in Macroeconomics

Sims solved the *estimation* problem of the mid-1900s: no more need for hundreds of atheoretical exclusion restrictions.

But the Lucas critique applies at the *identification* stage:

- VARs are reduced-form: Coefficients are convolutions of deep structural parameters and the prevailing policy rule. Change the rule, change the coefficients.
- To give structural meaning to a VAR, we still need identifying assumptions.
- Those assumptions embed implicit beliefs about how agents behave, and these beliefs may not survive a change in policy regime.

A pragmatic response: Think hard about identification to isolate genuinely exogenous shocks. Whether the Lucas critique prohibits SVARs remains debated. Top researchers use both structural VARs and DSGE models. Kilian & Lütkepohl (2019) provide a fantastic discussion of VARs and DSGE in Chapters 6 and 7.

Part 7

SVAR Tools

The dynamic effect of a structural shock

Moving Average Representation by Recursive Substitution

- Reconsider the VAR(1) but now go back in time:

$$y_t = (I + A_1 + A_1^2 + \cdots + A_1^j)c \; + \; A_1^{j+1} y_{t-(j+1)} \; + \; \sum_{i=0}^{j} A_1^i \, e_{t-i}$$

- **Stability** implies as $j \to \infty$:
  - $(I + A_1 + \cdots + A_1^j)c \;\longrightarrow\; (I - A_1)^{-1} c$ (constant converges)
  - $A_1^{j+1} y_{t-(j+1)} \;\longrightarrow\; 0$ (initial condition fades away)
- This gives the **moving average (MA(∞)) representation**:

$$y_t = v + \sum_{j=0}^{\infty} A_1^j \, e_{t-j}, \qquad v = (I - A_1)^{-1} \mu$$

Key Idea

Replacing $e_t = B_0^{-1} \varepsilon_t$ shows that the MA representation writes the VAR as a function of past structural shocks. We can use this to answer economic questions.

What Is an Impulse Response Function?

Impulse response function (IRF) of variable i to structural shock j at horizon h is:

$$\mathrm{IRF}_{ij}(h) = \frac{\partial y_{i,t+h}}{\partial \varepsilon_{jt}},$$

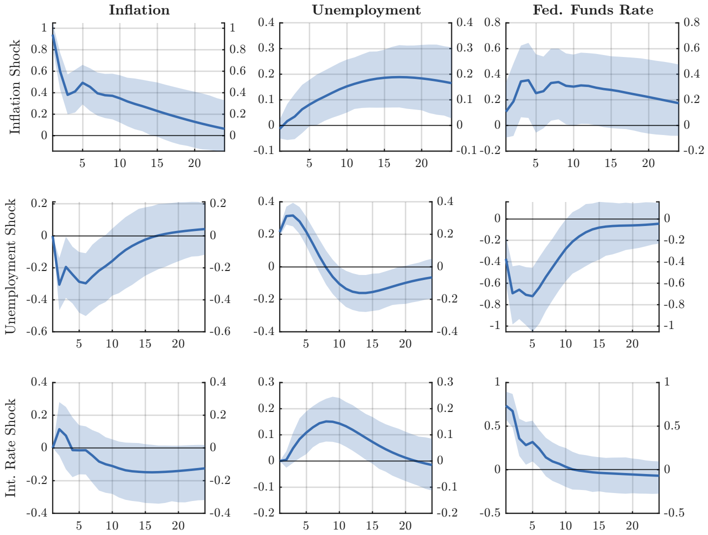
the partial derivative of variable i at time $t+h$ with respect to a unit shock to ε_j at time t.

- At $h = 0$: the *impact* response — immediate effect of the shock.
- At $h > 0$: the *dynamic* response — how the effect evolves over time.
- As $h \to \infty$: IRFs converge to zero for a stable VAR.

Key Idea

The IRF shows us how a variable reacts to a shock in another variable.

SW (2001): IRFs with 68% CI over 24 quarters.



Ex: Interpret the results

What Is a Forecast Error Variance Decomposition?

The **FEVD** of variable i at horizon h answers: what fraction of the h-step forecast error variance of y_i is attributable to structural shock j?

$$\text{FEVD}_{ij}(h) = \frac{\text{MSE}_{ij}(h)}{\sum_{j=1}^{n} \text{MSE}_{ij}(h)} \in [0, 1],$$

where $\text{MSE}_{ij}(h)$ is the contribution of shock j to the forecast error variance (FEV) of variable i.

- At each horizon h, the FEVDs for variable i sum to 1 across all shocks j.
- At $h = 1$: reflects only the *impact* effects of each shock.
- As $h \to \infty$: reflects the long-run importance of each shock.

Key Idea

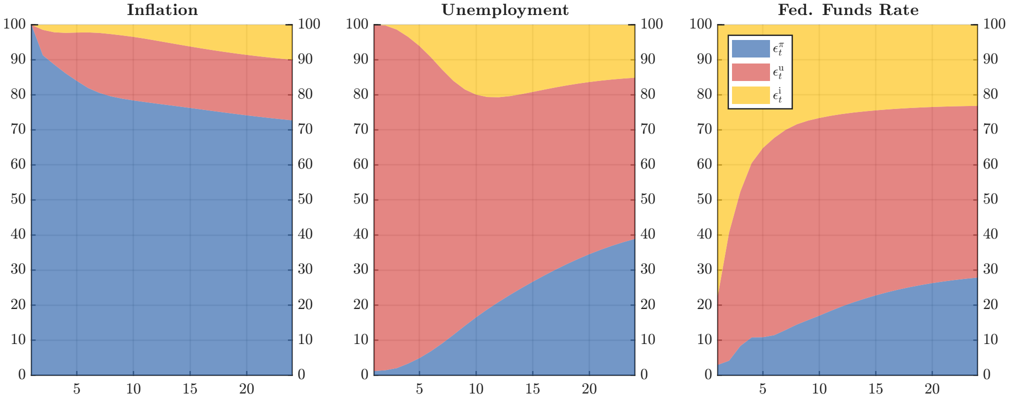
The FEVD tells us which structural shocks are the *most important drivers* of each variable.

SW (2001): FEVD of inflation (π) explained by each shock

Horizon	Inflation (π)	Unemployment (u)	Fed Funds Rate (R)
1	100	0	0
4	88	10	2
8	82	17	1
12	82	16	2

Ex: Interpret the results

SW (2001): FEVD Figure



Ex: Interpret the results

What Is a Historical Decomposition?

While the FEVD gives an *average* picture, the **Historical Decomposition (HD)** gives a *time-series* picture. The full decomposition is:

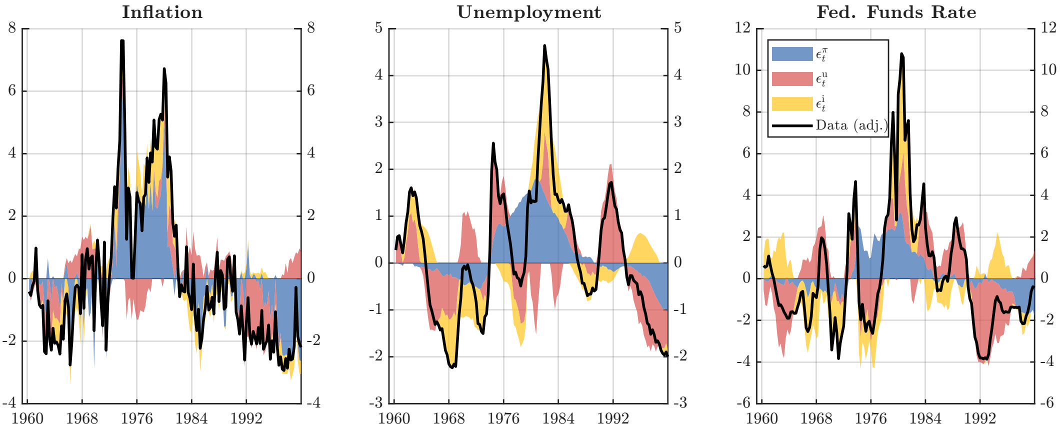
$$y_{it} = \underbrace{\mu_{it}}_{\text{deterministic (IC + intercept)}} + \underbrace{\sum_{j=1}^{n} \sum_{s=0}^{t-1} \mathrm{IRF}_{ij}(s)\, \hat{\varepsilon}_{j,t-s}}_{\text{stochastic: contribution of shock } j}$$

- **Deterministic component**: captures initial conditions and the intercept — often set aside (see Bergholt et al. (2026) for recent discussion).
- **Stochastic component**: shows how much each structural shock contributed to y_{it} at each point in time.

Key Idea

HDs tell us which structural shocks drove the variable movements over time.

Addition to SW (2001): HD Figure



Ex: Interpret the results

Summary

What Have We Learned?

Main takeaways from Day 1

Main Takeaways

1. **Why VARs?** Overcome ad hoc restrictions in pre-1980s econometrics. VARs are flexible, data driven, and allow for transparency in assumptions.
2. **Two types of VARs**
   2.1 **Reduced form**: System of equations where each variable depends on lags of others.
   2.2 **Structural**: System of equations where variables respond to structural shocks.
3. **Estimation**: OLS and CIs via bootstrap. However, large VARs require additional methods due to overparameterization (Classical Regularization or Bayes Methods).
4. **Specification**: Let theory guide empirics! Variables, lag length, and stability.
5. **Identification**: Assumptions are required to recover the SVAR. This is the most debated point in SVARs!
6. **SVAR Tools** allow us to examine the effects of **structural shocks**:
   6.1 **IRFs** trace the dynamic response to a structural shock
   6.2 **FEVDs** quantify each shock's contribution to variance
   6.3 **HDs** attribute historical fluctuations to specific shocks

Preview of Day 2

Short-run (zero) restrictions — the recursive approach in more detail, with applications to:

1. Oil markets (Kilian, 2009)
2. Inflation dynamics (Wong, 2015)

Thank you!